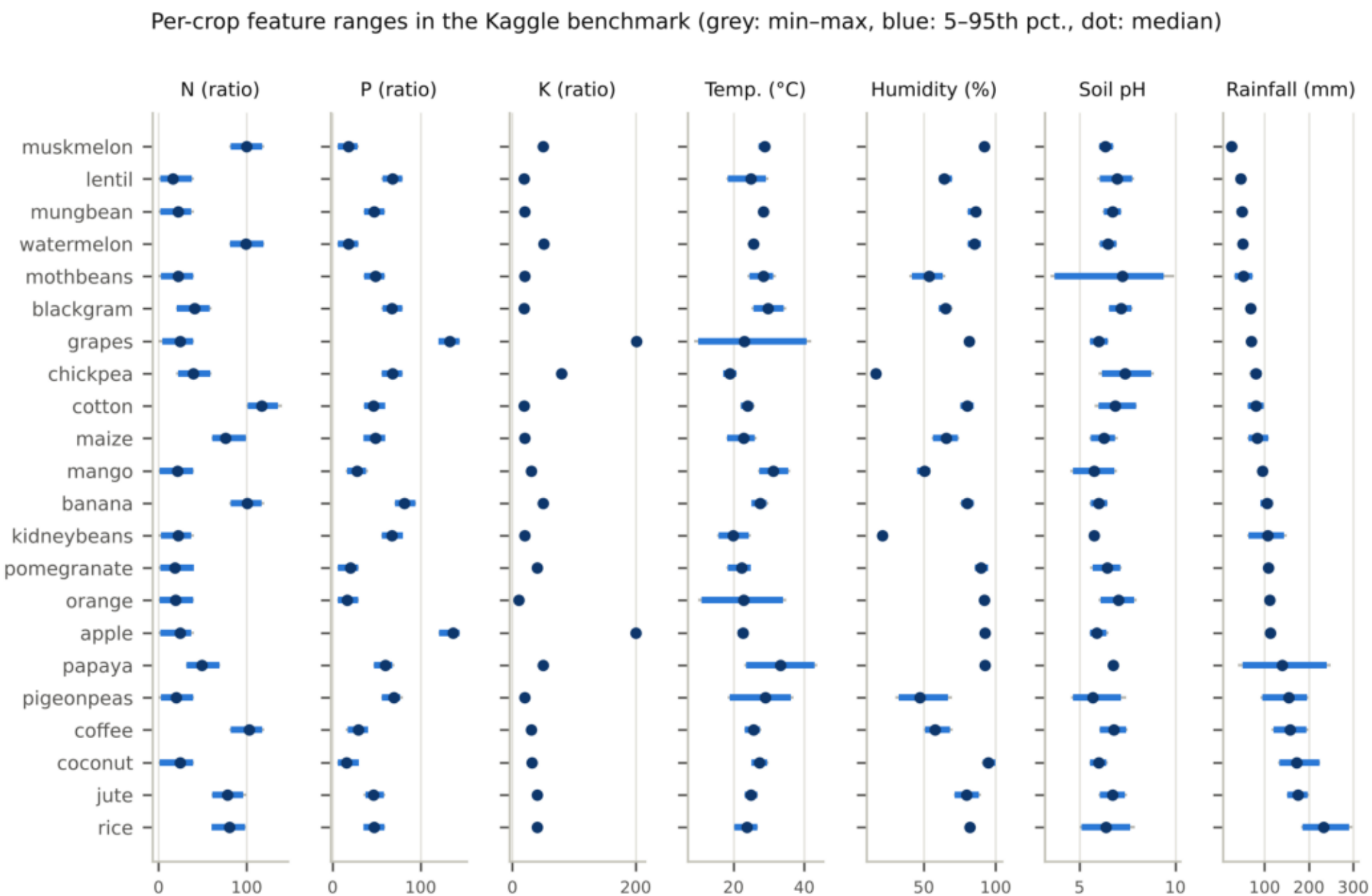
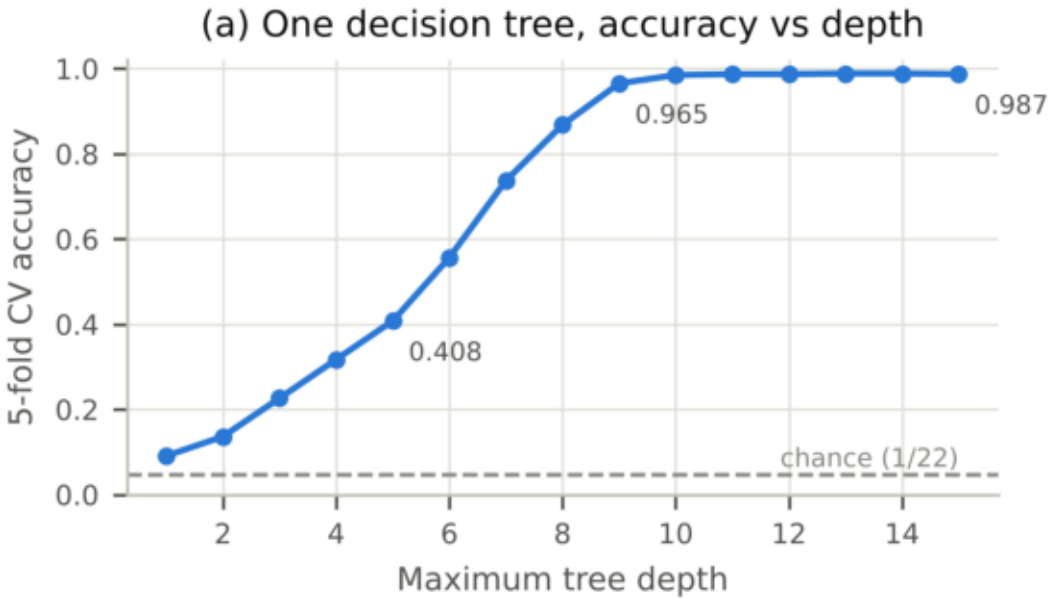


Fig 1. Per-crop feature ranges (Kaggle). Non-overlapping ranges make most crops separable by one or two features.



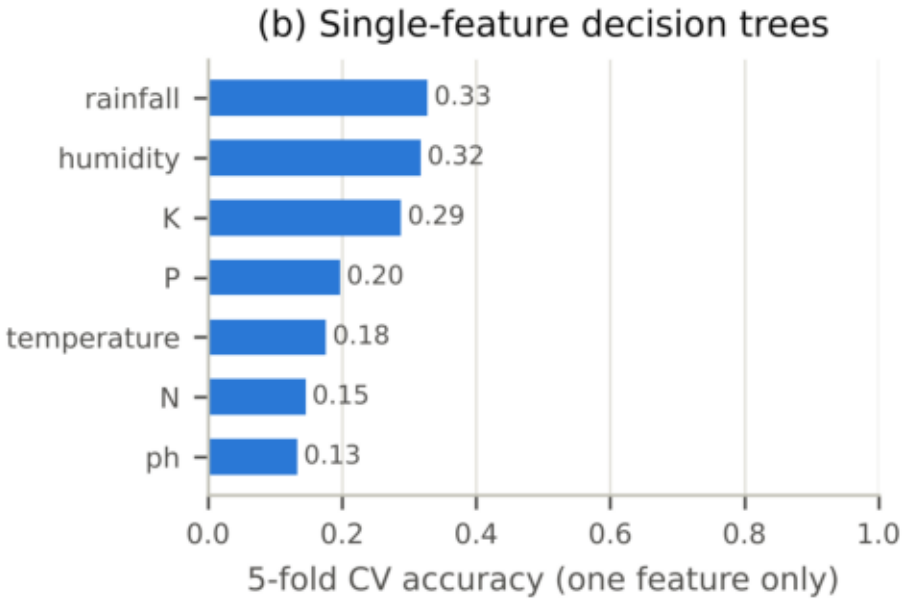


Fig 2. (a) One decision tree vs depth; (b) single-feature trees. Humidity+rainfall alone: 76.9% CV accuracy.

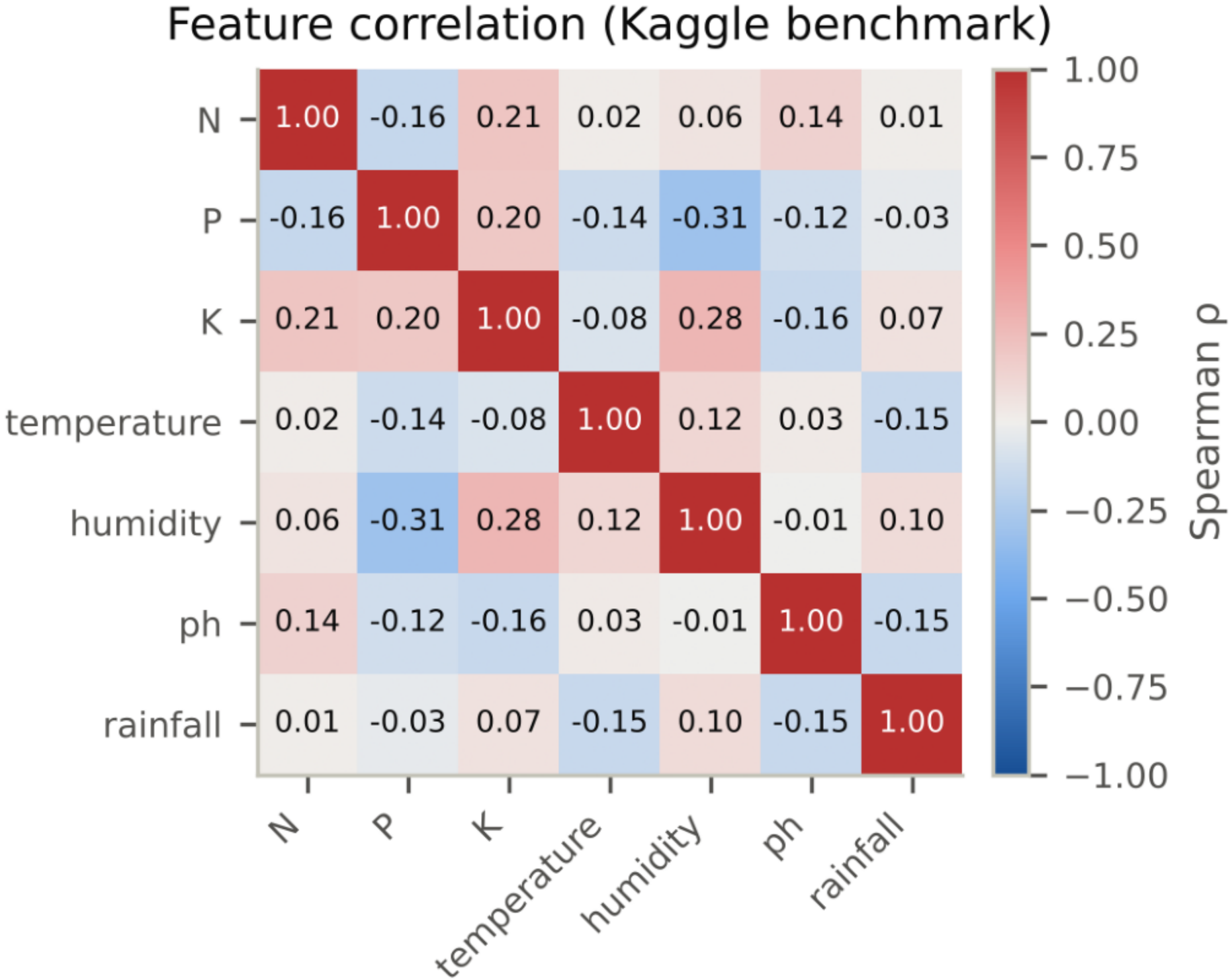


Fig 3. Spearman correlation between features.

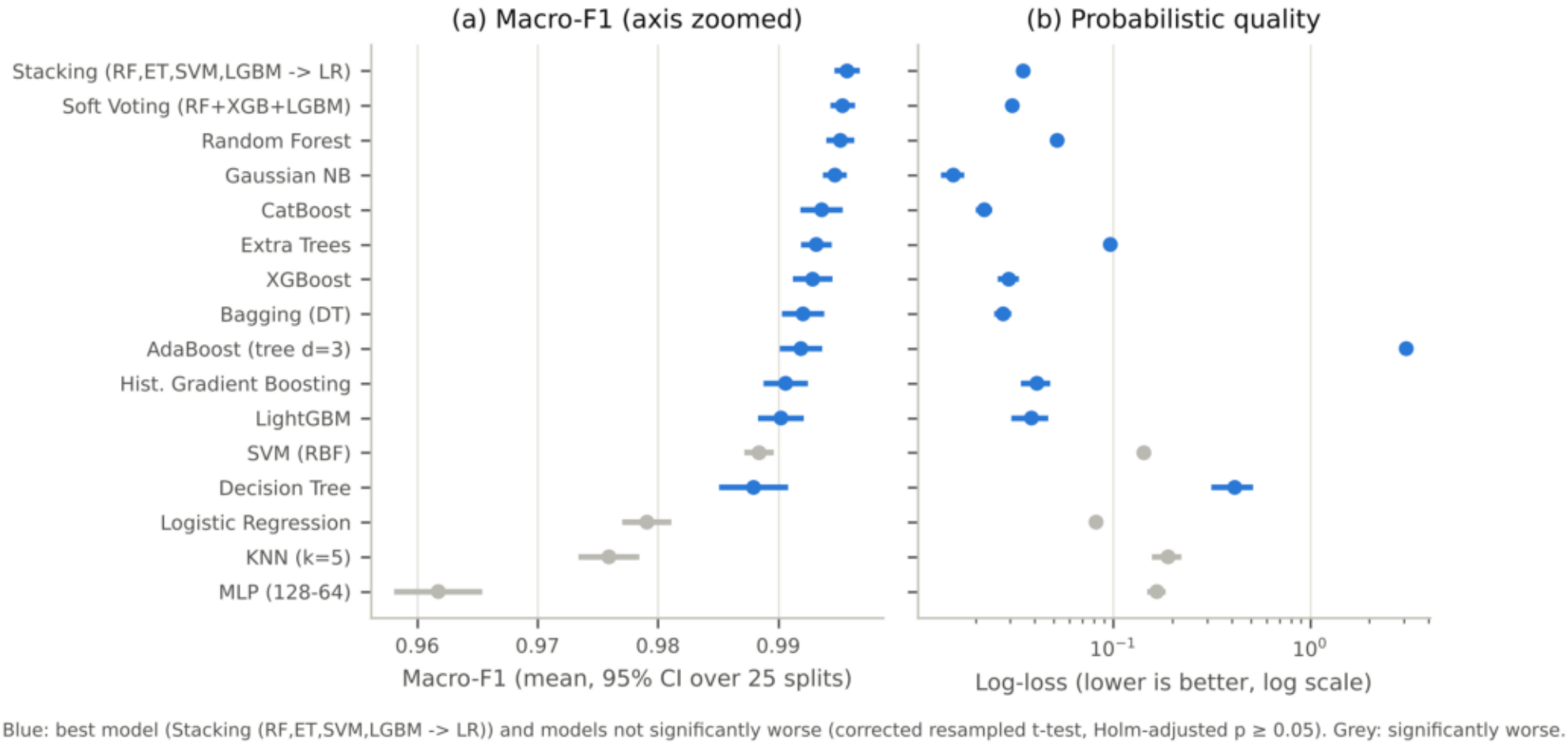


Fig 4. 16 models, 5×5 repeated stratified CV. 11/16 not significantly worse than the best (Holm-corrected).

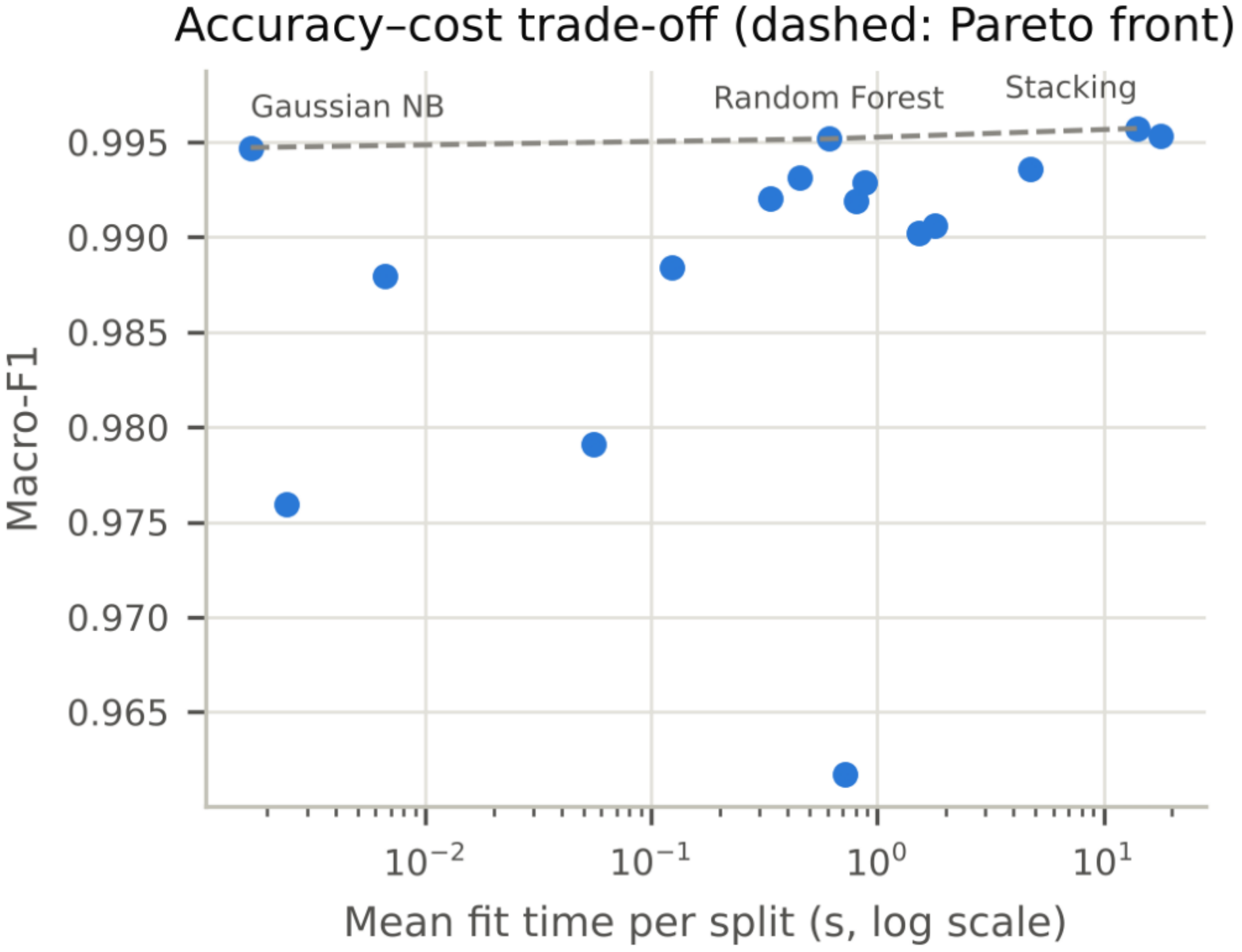


Fig 5. Fit time vs macro-F1; Gaussian NB is on the Pareto front at ~0 s.

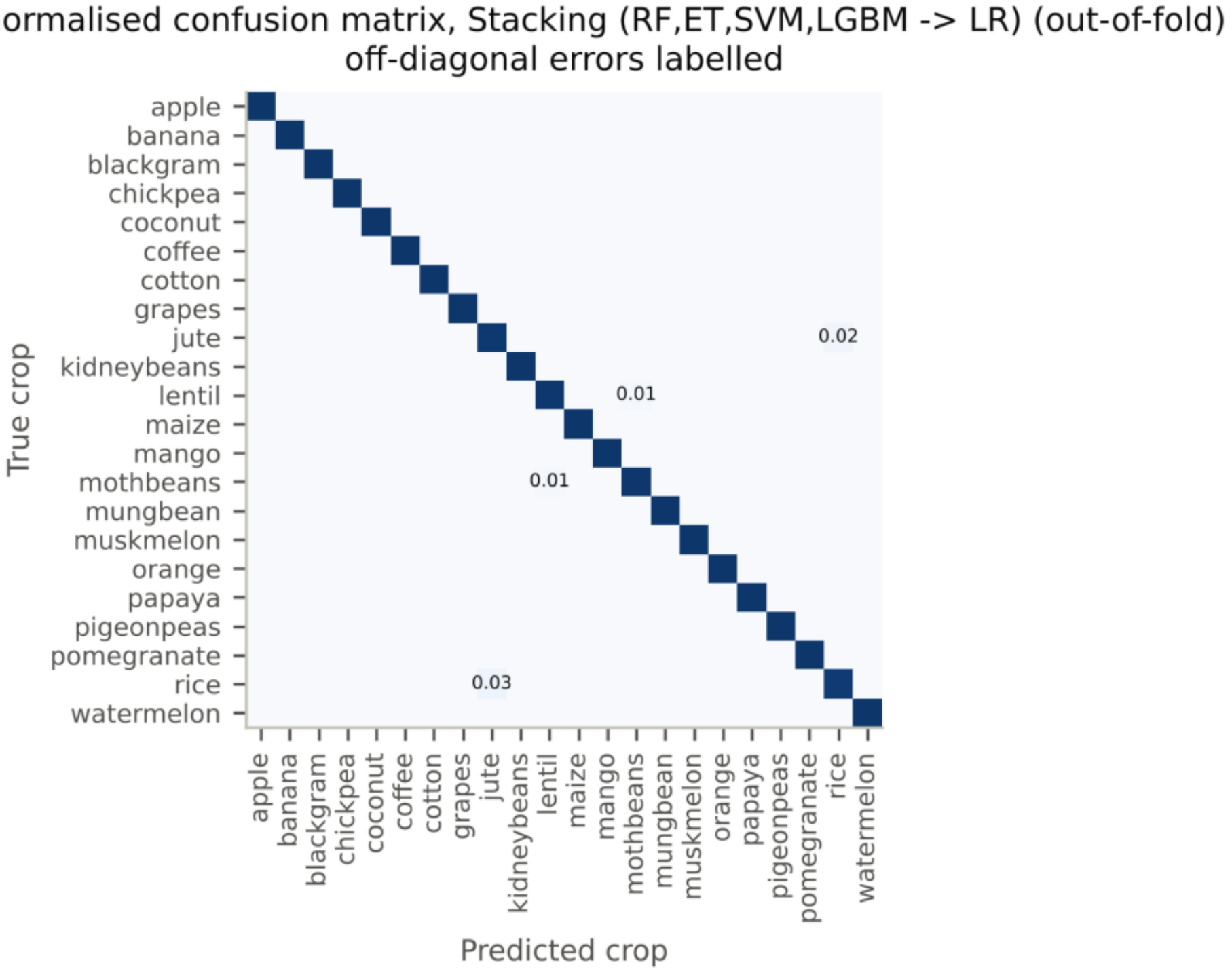


Fig 6. Confusion matrix of the best model; residual errors are rice/jute and lentil/mothbeans.

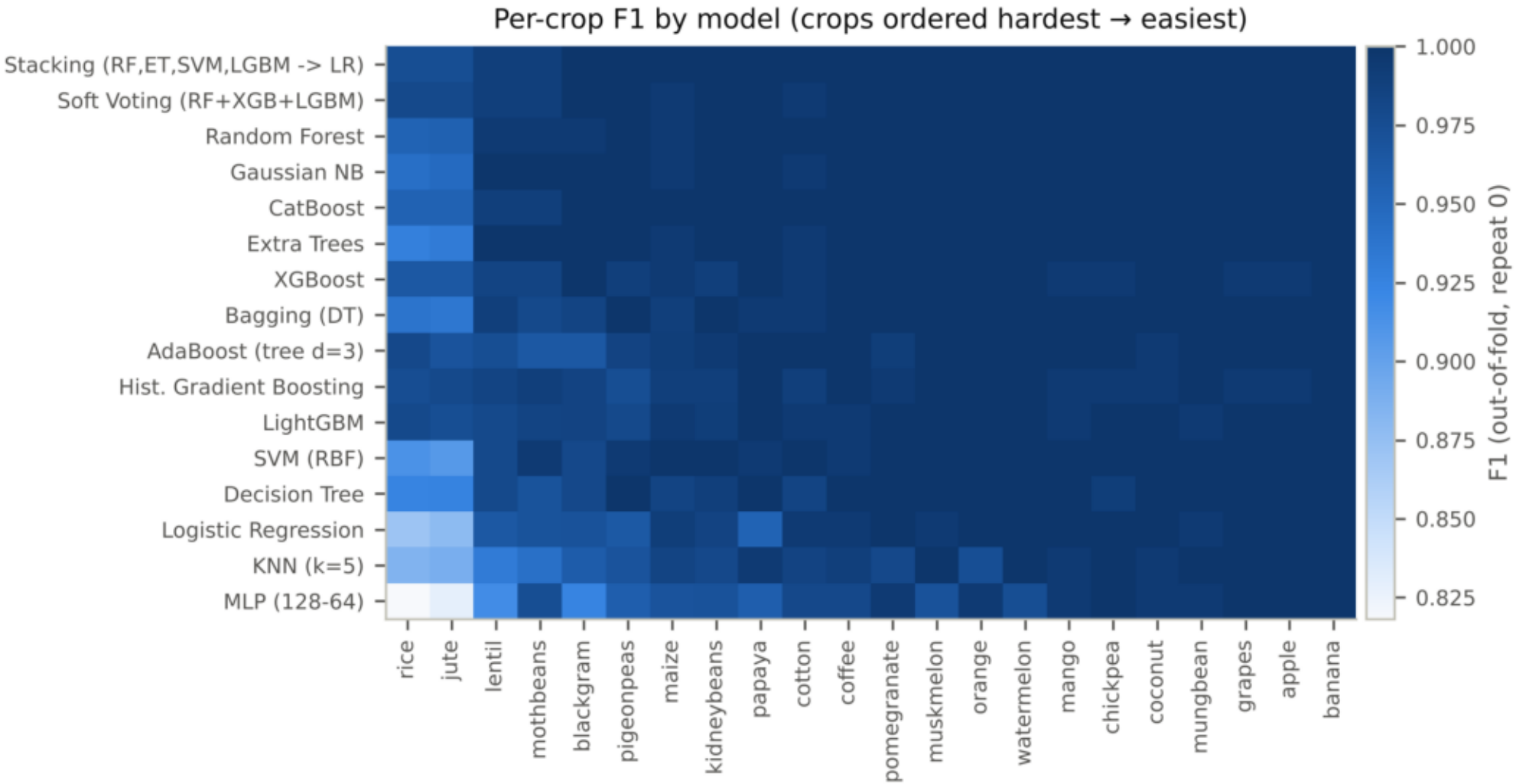


Fig 7. Per-crop F1 for every model.

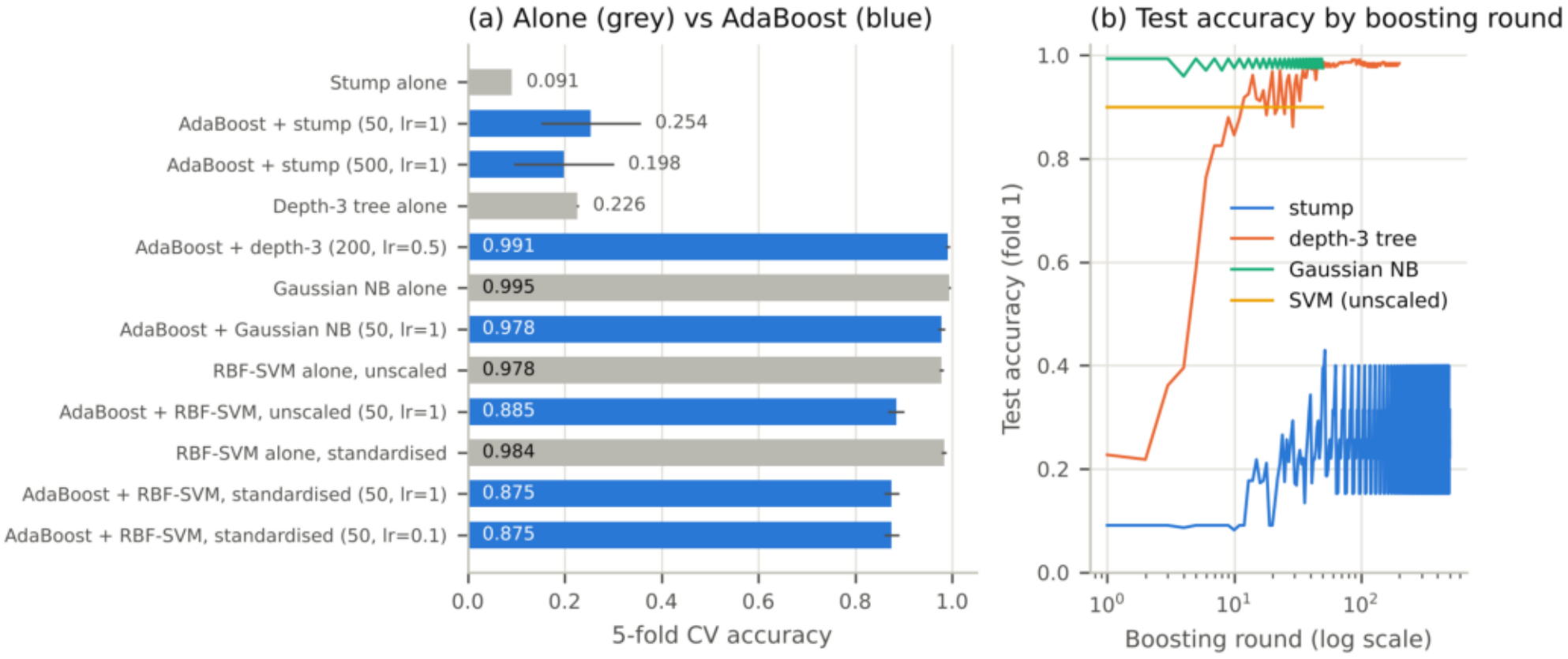


Fig 8. AdaBoost with default stumps collapses to 20-25%; boosting SVM/NB lowers their accuracy.

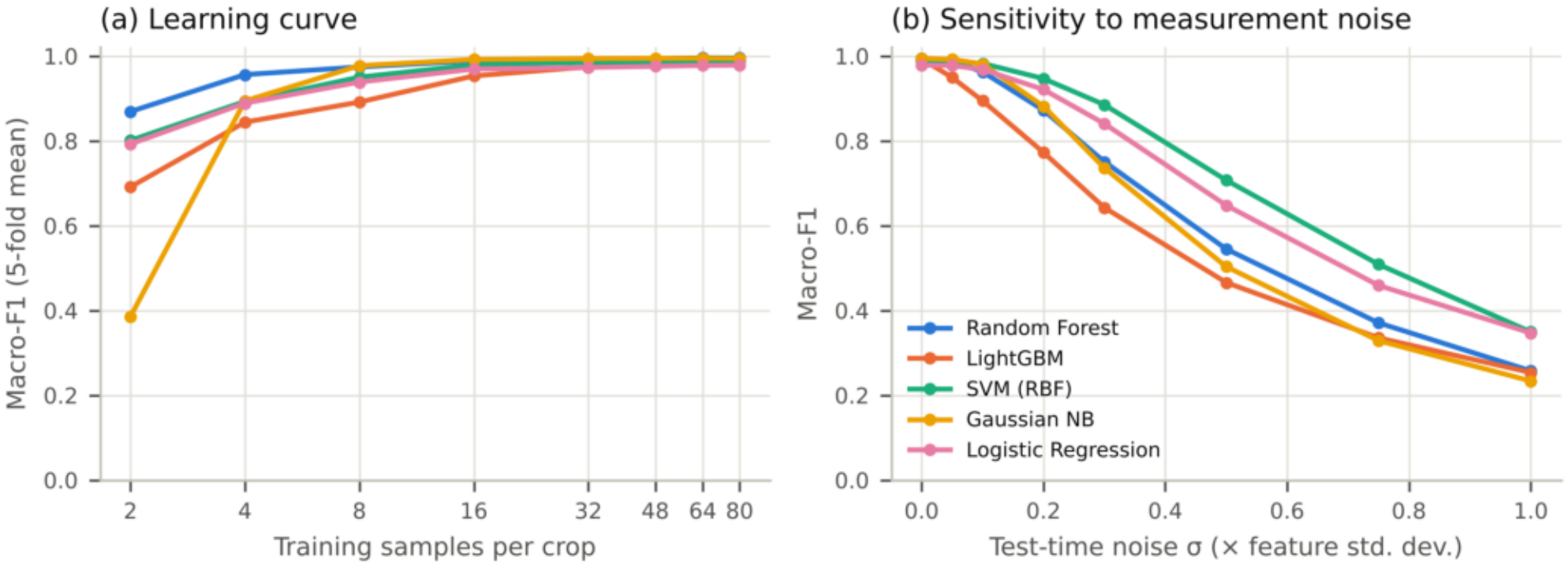


Fig 9. (a) RF reaches 0.97 macro-F1 with 8 samples per crop; (b) ranking changes under measurement noise.

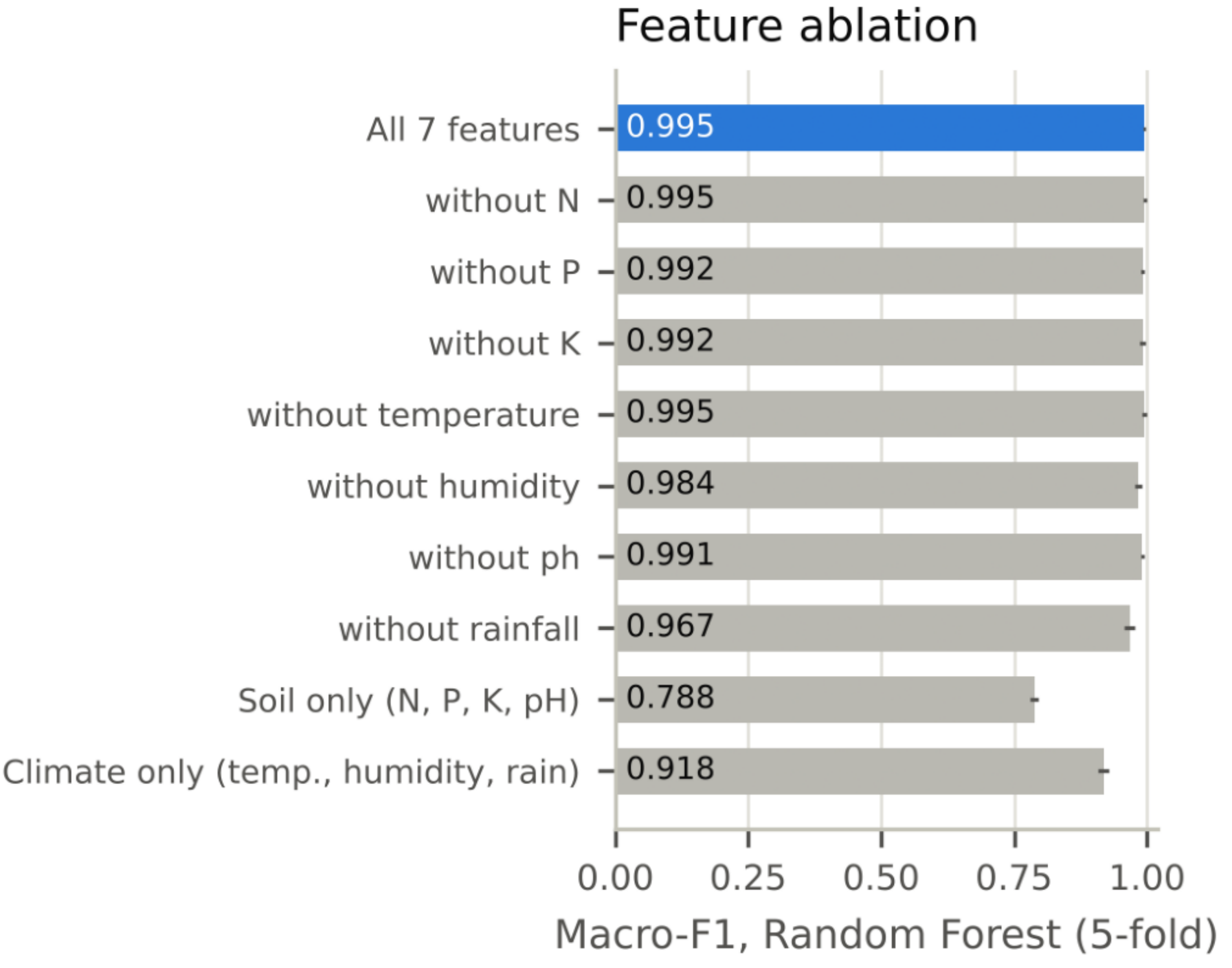


Fig 10. Climate-only features beat soil-only (0.918 vs 0.788 macro-F1).

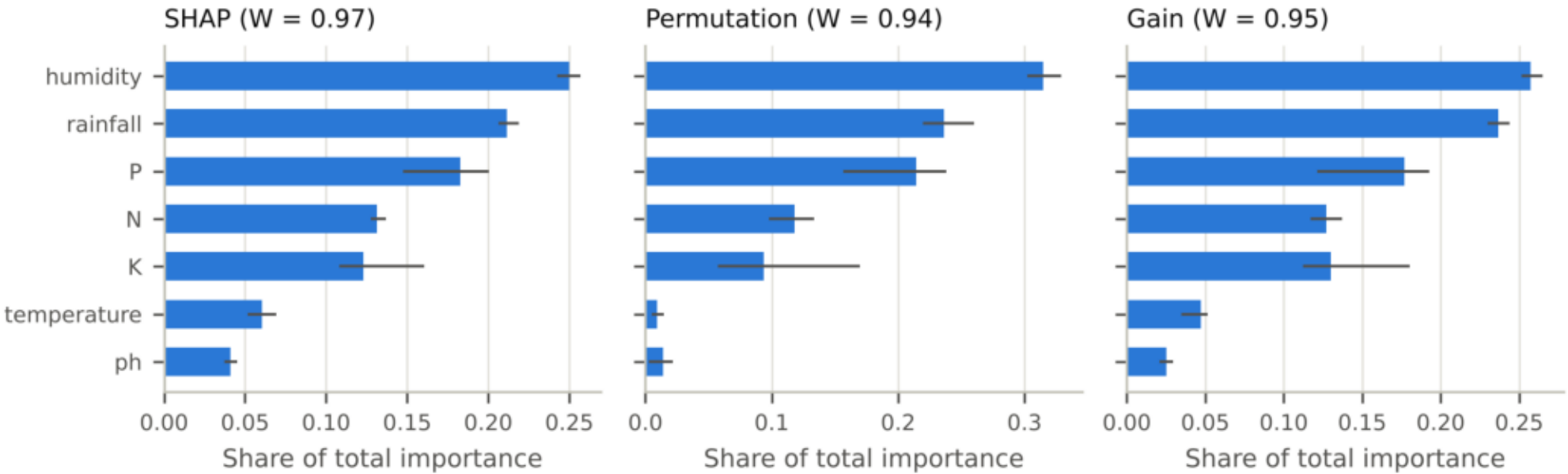


Fig 11. SHAP, permutation and gain importance with split ranges and Kendall's W.

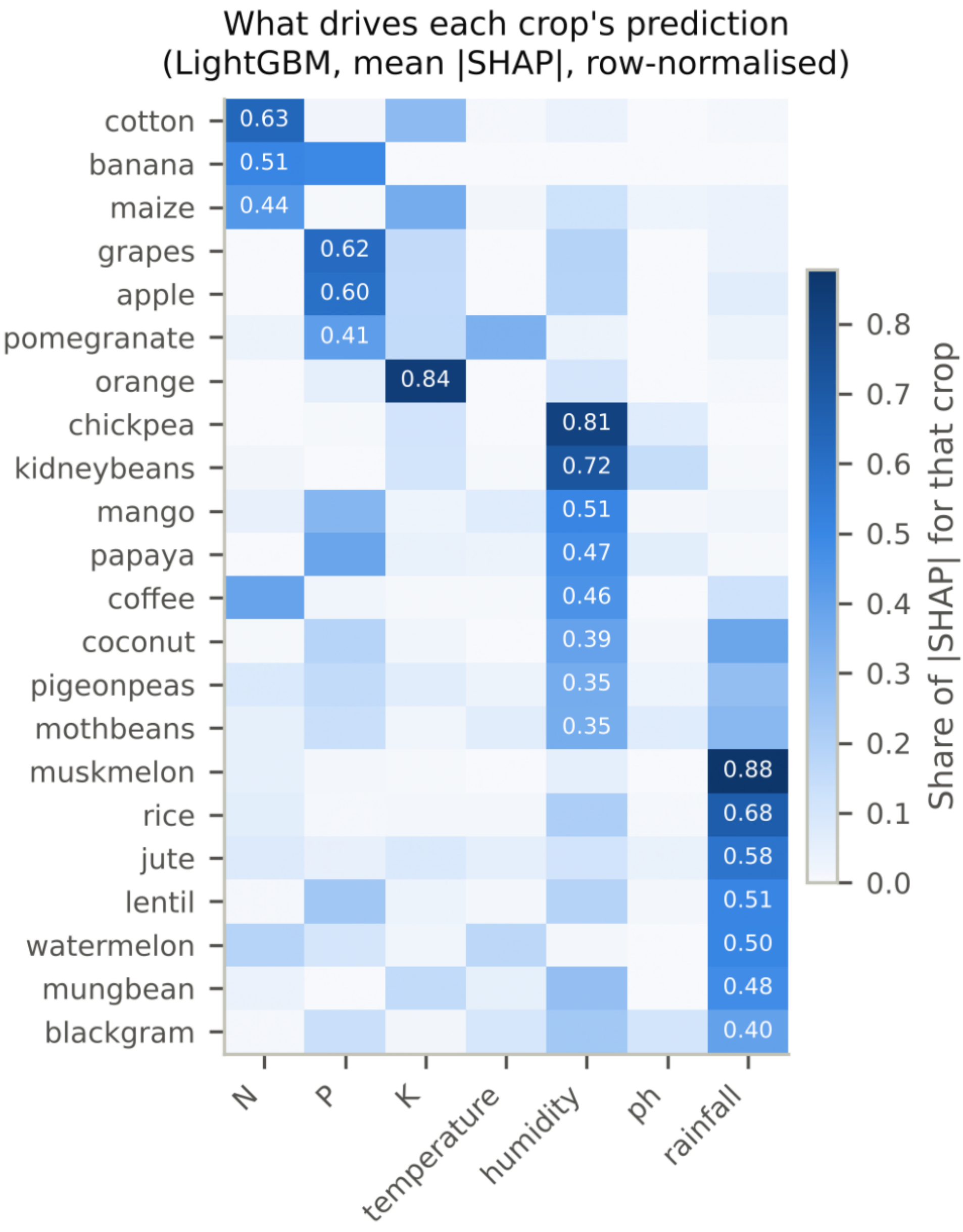


Fig 12. Feature driving each crop's prediction (row-normalised mean |SHAP|).

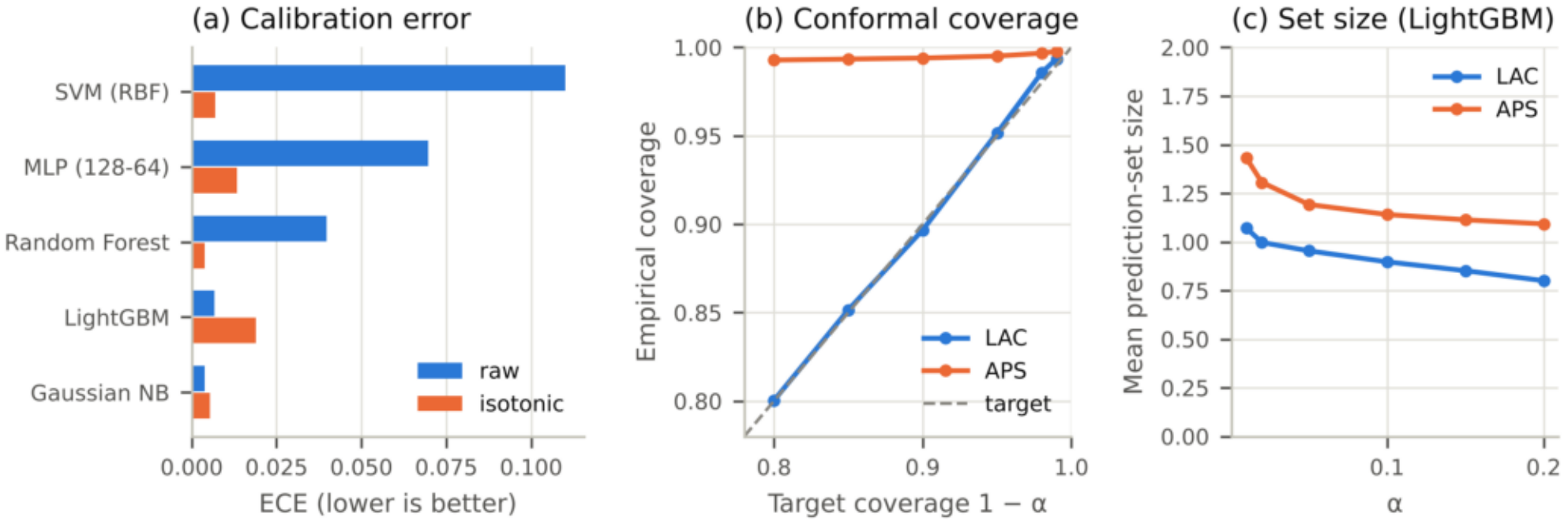


Fig 13. (a) ECE before/after isotonic; (b,c) conformal coverage and set size.